Multimodal Urgency Detection Using Prosodic and Conversational Cues

INDIVIDUAL THESIS

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ABSTRACT

Emergency call centres in Sri Lanka must judge the urgency of incoming reports within seconds, across Sinhala, Tamil, English, and code-mixed speech, often from transcript content alone that cannot reliably distinguish a life-threatening call from a routine one. This individual component of a four-part group research project addresses that gap by building a multimodal urgency classifier that combines acoustic-prosodic and textual/conversational features, and by independently testing whether performance measured during development actually transfers to authentic emergency recordings - a question prior urgency-classification research has rarely asked directly. The classifier was developed on a 267-call simulated corpus of role-played fire and flood emergency calls under five-fold cross-validation, comparing unimodal and multimodal (early- and late-fusion) formulations, and was then evaluated independently against 59 authentic recordings from Sri Lanka's Disaster Management Centre, which played no part in development. Early fusion reached the best development-corpus macro-F1 (0.630) but fell to 0.326 on the independent set, while text-only features alone improved from 0.397 to 0.449 macro-F1 - a reversal only visible because of the external evaluation. A domain-shift analysis using Cohen's d showed acoustic features shifted considerably more between the two corpora than textual features, and a targeted audio-augmentation experiment found that simple telephony-noise and bandwidth transformations do not by themselves explain that shift. Guided by the independent evaluation, a late-fusion formulation was selected as final, achieving the strongest High-urgency recall (46.1%) among the configurations tested. The individual contribution of this work is this independently validated comparison itself: a transparent, evidence-based account of where multimodal fusion helps an urgency classifier transfer to authentic recordings, and where it does not, intended to inform the urgency-classification input to the group's four-component emergency-response system.

Keywords: emergency call triage, urgency classification, multimodal fusion, domain shift, code-mixed speech

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TABLE OF CONTENTS

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 1

1.3 Problem statement 2

1.4 Research gap 3

1.5 Aim and objectives 3

1.6 Research questions or hypotheses 4

1.7 Scope limitations and assumptions 4

1.8 Individual contribution 5

1.9 Thesis structure 6

2 LITERATURE REVIEW AND RELATED WORK 7

2.1 Review method and scope 7

2.2 Domain and theoretical foundations 7

2.3 Related research 9

2.4 Existing systems products and techniques 12

2.5 Gap synthesis and conceptual framework 14

3 RESEARCH METHODOLOGY 16

3.1 Research design 16

3.2 Data participants or study materials 16

3.3 Experimental or study procedure 18

3.4 Measures and evaluation metrics 19

3.5 Data analysis 19

3.6 Ethics privacy security and responsible AI 20

3.7 Reproducibility and research artefacts 21

4 PRODUCT AND SOLUTION DESIGN 23
4.1 Requirements 23
4.2 Component architecture within the complete solution 24
4.3 Design alternatives and decisions 25
4.4 Detailed design 26
4.5 Standards and professional practice 27
5 IMPLEMENTATION INTEGRATION AND TESTING 29
5.1 Development environment and technology stack 29
5.2 Component implementation 29
5.3 Integration with the other components 31
5.4 Verification and software testing 32
5.5 Deployment and operation 32
6 RESULTS AND VALIDATION 34
6.1 Research results 34
6.2 Product and technical results 39
6.3 User or stakeholder validation 39
6.4 Integration results 40
6.5 Summary of findings 40
7 DISCUSSION 42
7.1 Interpretation of findings 42
7.2 Comparison with literature baselines and existing solutions 43
7.3 Research contribution 44
7.4 Product contribution 44
7.5 Limitations and threats to validity 44
7.6 Implications 45
8 COMMERCIALISATION AND SUSTAINABILITY 47

8.1 Value proposition and target users 47
8.2 Market and competitive position 47
8.3 Business or sustainability model 47
8.4 Scalability intellectual property and compliance 48
8.5 Contribution to the Sustainable Development Goals 48
9 CONCLUSION AND FUTURE DIRECTIONS 50
9.1 Conclusion 50
9.2 Achievement of objectives 50
9.3 Individual research and product contributions 51
9.4 Future directions 51
REFERENCES 53
GLOSSARY 56
APPENDICES 57
Appendix A Research instruments or extended results 57
Reproduction commands 57
Worked example: one DMC call through the full pipeline 59
Appendix B Academic integrity and AI use disclosure 61

LIST OF FIGURES

Figure 3.1: Urgency-class distribution, simulated vs. DMC evaluation corpus 17

Figure 4.1: Multimodal urgency-classification pipeline — late fusion of acoustic and text/context branches 25

Figure 6.1: Macro-F1 and accuracy by formulation, simulated vs. DMC evaluation 35

Figure 6.2: Per-class recall of the final late-fusion model, simulated vs. DMC 35

Figure 6.3: Confusion matrix of the final late-fusion model, DMC Full 36

Figure 6.4: Top 15 features by |Cohen's d|, simulated vs. DMC 37

Figure 6.5: PCA projection of the 91 acoustic features, simulated vs. DMC Full 38

LIST OF TABLES

Table 2.1: Comparison of this component against related research and techniques 14

Table 4.1: Requirements traceability 23

Table A.1: Confusion matrix, locked late-fusion model, DMC Full (59 calls) 57

Table A.2: Acoustic feature-subset ablation, simulated corpus, 5-fold CV 58

Table A.3: Domain-shift ranking by |Cohen's d|, ranks 16-30 59

Table A.4: Feature-selection ablation, simulated development corpus 60

LIST OF ABBREVIATIONS

Abbreviation	Meaning
ASR	Automatic Speech Recognition
CI	Confidence Interval
CV	Cross-Validation
DMC	Disaster Management Centre
F1 / Macro-F1	F1 score / macro-averaged F1 score across classes
MFCC	Mel-Frequency Cepstral Coefficient
RF	Random Forest
SDG	Sustainable Development Goal
SMOTE	Synthetic Minority Over-sampling Technique

1 INTRODUCTION

1.1 Overall project context

The Multilingual Multi-Channel Emergency Intake and Triage Pipeline for Sri Lanka is a four-component group project developed to support faster, more informed emergency response in Sri Lanka. Emergency response in multilingual, resource-constrained settings depends on dispatchers and responders being able to quickly judge which incoming reports need immediate action, while processing information that arrives through several different channels and in several languages. The group project addresses this by combining four components, each targeting a different part of the problem: (1) multimodal urgency detection from emergency call audio and transcripts, which is the subject of this individual thesis; (2) a hybrid triage agent component that restricts large language model use to stages where rule-based extraction is demonstrably brittle, wraps every generative call in an evidence-grounding guard, and assigns dispatch priority using the Medical Priority Dispatch System; (3) a closed-loop SMS dialogue component that decides, turn by turn, which single missing field to request from the reporter, using an auditable finite-state clarification policy rather than a generative selector; and (4) a multi-source incident fusion component that brings the outputs of the other components together into a single, structured incident picture for a dispatcher or response agency to act on.

These four components are designed to be used together rather than in isolation. A single incoming report, whether a phone call or a text message, can be processed by the relevant component, and its structured output - including an urgency judgement, in the case of the component described in this thesis - is intended to feed into the shared incident-fusion layer. The combined system's objective is to reduce the time between a report being received and an appropriately prioritised response being dispatched, and to make that prioritisation more consistent than relying on an individual call-taker's judgement alone under time pressure.

1.2 Individual component context

This thesis covers the multimodal urgency detection component: a classifier that assigns a High, Medium, or Low urgency label to an emergency call by combining

two independent sources of evidence from the call itself - acoustic-prosodic features extracted directly from the caller's voice, and textual/conversational features extracted from an automatic transcript of the call. The component operates in the multilingual context of Sri Lanka, where callers communicate in Sinhala, Tamil, English, or a code-mixed combination of these within a single call.

The component was developed in two stages. In the first stage, it was built and validated on a 267-call simulated corpus of role-played fire and flood emergency calls, constructed specifically for this project. In the second stage, and central to this thesis, the component was evaluated independently against 59 authentic recordings obtained from Sri Lanka's Disaster Management Centre (DMC) - recordings the classifier never saw during development - to test whether performance measured on the simulated corpus actually transfers to real emergency calls. That independent evaluation, and what it reveals about the gap between simulated and authentic performance, is the central empirical contribution of this individual thesis.

1.3 Problem statement

Call-taking staff at an emergency response centre must judge, within seconds and often from a brief and emotionally charged exchange, how urgently a given report needs to be acted on. Transcript content alone is frequently not enough to make this judgement reliably: a caller describing an immediately life-threatening situation and a caller asking a routine informational question can use much of the same vocabulary, while sounding completely different in how they say it. A classifier that relies only on what was said, and ignores how it was said, is therefore working with an incomplete signal.

At the same time, a classifier that is only ever developed and tested on simulated or role-played emergency calls - which is typically the only data realistically available during development, given the practical and ethical difficulty of collecting labelled real emergency audio - cannot be assumed to generalise to authentic calls. Authentic recordings differ from simulated ones in recording conditions, caller behaviour under genuine distress, and the real-world mix of incident types and urgency levels. The problem this thesis addresses is therefore two-fold: first, whether combining acoustic and textual evidence can produce a more reliable urgency classifier than either source

alone; and second, and more critically, whether a classifier that performs well on simulated development data actually performs well on independent, authentic emergency recordings, and if not, why not.

1.4 Research gap

Existing work on classifying the urgency of emergency-related communication has generally reported performance on a single, internally consistent corpus - either entirely simulated or entirely real - rather than developing a classifier on one and independently evaluating it on the other [8], [9], [10]. This leaves open exactly the question that matters for real deployment: does a result that looks strong during development actually transfer to authentic recordings collected outside the development process? Separately, speech-emotion and paraverbal research has established that acoustic correlates of stress and arousal carry information that transcript content alone does not capture [4], but prior emergency-call classification work has generally relied on transcript content, leaving the acoustic channel comparatively underused despite its documented robustness to transcription failure.

The multilingual, code-mixed character of Sri Lankan emergency communication adds a further gap: cross-lingual and code-switching research shows that findings and tooling built for monolingual settings cannot simply be assumed to hold when speakers mix Sinhala, Tamil, and English within a single utterance [1], [2], [3]. No prior study, to the knowledge gathered during this research, has combined these threads - multimodal (acoustic and textual) urgency classification, a multilingual and code-mixed Sri Lankan setting, and a quantitative, independently validated comparison of simulated-corpus performance against authentic-recording performance - for emergency-call triage specifically. This thesis is positioned to close that combined gap.

1.5 Aim and objectives

Aim: To design, build, and independently validate a multimodal urgency classifier for Sri Lankan multilingual emergency calls, and to quantify how far simulated-corpus performance can be trusted as an estimate of performance on authentic recordings.

Objectives:

1. To design and extract a set of acoustic-prosodic and textual/conversational features suited to the multilingual, code-mixed Sri Lankan emergency-call setting.

2. To build and compare unimodal (acoustic-only, text-only) and multimodal (early-fusion and late-fusion) urgency classifiers on a simulated development corpus.

3. To independently evaluate every classifier formulation against authentic recordings collected from Sri Lanka's Disaster Management Centre, and to use that external evidence - rather than development-set performance alone - to select a final classifier formulation.

4. To quantify, using a domain-shift analysis, which feature families differ most between the simulated and authentic recordings, and to test directly whether simple telephony-channel differences explain that shift.

5. To integrate the resulting urgency-classification output with the other three components of the group project, so that it can inform the shared incident-fusion layer.

1.6 Research questions or hypotheses

RQ1. How effectively can acoustic-prosodic and textual/conversational information, individually, classify the urgency of an emergency call?

RQ2. Does multimodal fusion improve urgency classification over either modality alone, and does the fusion strategy used - early fusion versus late fusion - matter?

RQ3. How well does a classifier developed on a simulated emergency-call corpus generalise to independent, authentic recordings?

RQ4. What measurable factors help explain the observed difference between simulated-corpus and authentic-recording performance?

1.7 Scope limitations and assumptions

Scope. The component covers emergency calls in Sinhala, Tamil, English, and their code-mixed combinations, and classifies each call into one of three urgency levels: High, Medium, or Low. The simulated development corpus was constructed around fire and flood scenarios specifically, and the independent DMC validation set, while not restricted to those two incident types, is drawn predominantly from flood-related

operations. Model development used a Random Forest classifier under five-fold cross-validation on the simulated corpus, with four fusion formulations compared (text-only, acoustic-only, early fusion, late fusion).

Limitations and assumptions. The DMC recordings, while genuinely independent of the simulated corpus, directly informed the choice of the final fusion formulation; this thesis is therefore explicit throughout that the reported DMC evaluation is an external validation that guided model selection, not an untouched, unbiased test-set estimate of generalisation. Both the simulated corpus (267 calls) and the DMC set (59 calls) are modest in size for a three-class problem, so per-class estimates - particularly on the independent set - carry meaningful statistical uncertainty. The domain-shift analysis in this thesis establishes association between feature-family shift and transfer performance, not causation. The urgency labels themselves were assigned using an operational, but not formally standardised, definition obtained from DMC staff (High: immediate threat to life; Low: informational or non-emergency; Medium: a present but non-immediate threat where help is needed but not urgently), since DMC does not maintain a formally documented urgency taxonomy. This classifier makes no claim to offer a clinical or safety guarantee; it is reported strictly as a decision-support signal.

1.8 Individual contribution

This thesis makes the following individual contributions:

1. A systematic comparison of unimodal and multimodal (early- and late-fusion) urgency classifiers for emergency calls in a multilingual, code-mixed setting.

2. An independent external validation against 59 authentic DMC recordings - distinct from the simulated development corpus in source, collection method, and time period - used directly to select the final fusion formulation rather than reported alongside an unrelated final choice.

3. A quantitative domain-shift analysis using Cohen's d, backed by a targeted audio-augmentation experiment, identifying which feature families differ most between the simulated and authentic distributions, and showing that simple telephony-noise and bandwidth transformations do not by themselves account for that difference.

4. A transparent, evidence-based account of exactly where the multimodal reformulation improves on text-only performance, and where it does not - including the finding that text-only features alone transfer better, by some metrics, than the early-fusion model that performed best during development.

5. Integration of the urgency-classification output into the shared four-component system, providing a structured urgency signal to the multi-source incident-fusion layer.

1.9 Thesis structure

The remainder of this thesis is organised as follows. Chapter 2 reviews related work on emergency-call and text-based urgency classification, speech-based affect and urgency cues, multilingual and code-mixed language processing, and domain shift, and synthesises the gap this thesis addresses. Chapter 3 describes the research methodology, including the datasets, feature engineering, classifier design, and evaluation protocol, together with the ethical considerations around using real emergency call recordings. Chapter 4 covers the design of the component and its place within the overall four-component system. Chapter 5 describes the implementation, integration with the other components, and testing approach. Chapter 6 presents the results, including the simulated-corpus comparison, the independent DMC evaluation, and the domain-shift analysis. Chapter 7 discusses these results in relation to the research questions and existing literature. Chapter 8 considers the commercialisation and sustainability aspects of the component. Chapter 9 concludes the thesis and outlines directions for future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review covers academic literature relevant to four themes that together define the problem addressed by this component: (i) classifying the urgency of emergency-related communication, by text, speech, or both; (ii) speech-based affect and paraverbal cues as a signal distinct from lexical content; (iii) multilingual and code-mixed natural language processing, including automatic speech recognition, relevant to the Sinhala-Tamil-English setting; and (iv) domain shift between data used for model development and data encountered in deployment. Sources were located through Google Scholar, arXiv, and ACL/IEEE digital libraries, using combinations of the search terms “urgency classification”, “emergency call classification”, “speech emotion recognition”, “code-switching”, “multilingual NLP Sinhala”, and “domain shift” / “dataset bias”. Sources were included where they were peer-reviewed, from a reputable conference or journal, or a widely cited pre-print, and directly relevant to at least one of the four themes above; general machine-learning methodology papers not specific to these themes were excluded unless they underpin a technique used directly in this component (for example, SMOTE for class imbalance). This is a targeted, thematic review rather than an exhaustive systematic review, scoped to the specific combination of problems this component addresses.

2.2 Domain and theoretical foundations

Emergency dispatch and call-taking is fundamentally a triage problem under uncertainty: a call-taker must allocate limited, time-critical attention across incoming reports before the true severity of each one is fully known. This operational reality motivates urgency classification as a decision-support problem rather than a diagnostic one - the classifier's role is to help prioritise attention, not to replace the human judgement involved in dispatch.

Speech emotion and paraverbal research provides the theoretical basis for using acoustic features alongside text. Vocal cues such as pitch, energy, speaking rate, and voice quality are understood to carry information about a speaker's arousal and affective state largely independently of the words used, which is why paraverbal

representations such as mel-frequency cepstral coefficients (MFCCs) are a standard basis for speech emotion recognition [4]. This theoretical separation between what is said and how it is said is the foundation for treating acoustic and textual features as complementary, rather than redundant, sources of evidence. In the broader multimodal-learning literature, this complementarity is usually realised through one of a small number of recognised fusion strategies - early (feature-level) fusion, late (decision-level) fusion, and hybrid intermediate schemes - each trading off differently between how much cross-modal interaction the model can learn and how robust it remains when one modality is noisy or degraded [19]; this thesis's own comparison of early- and late-fusion formulations (Section 3.4, Section 6.1) sits squarely within that taxonomy. This complementarity is reinforced empirically by work establishing standard baselines for deep-learning-based multimodal sentiment classification across fixed dataset splits, which found that naive feature concatenation is consistently outperformed by architectures that model cross-modal interaction explicitly [22].

Multilingual and code-switching research provides the second theoretical foundation. Cross-lingual representation learning shows that a single model can be trained to generalise across languages with limited per-language supervision [1], and large-scale weakly supervised speech recognition extends this to transcription [2]. However, the code-switching literature establishes that intra-sentential mixing of languages - as occurs routinely in Sinhala-Tamil-English emergency calls - is not simply noisy monolingual data, but a distinct linguistic phenomenon requiring explicit treatment in both language identification and downstream feature design [3].

The third theoretical foundation is domain shift: the well-established finding that a model's performance on its own training distribution does not predict its performance on a differently sourced dataset, sometimes termed dataset bias [5]. This concept is the theoretical basis for this thesis's central methodological decision - to treat simulated-corpus performance as provisional until checked against an independently sourced, authentic dataset - and for the quantitative domain-shift analysis presented in Chapter 6.

2.3 Related research

Emergency-call and text-based urgency classification. Machine-learning classification of emergency-call transcripts has been shown feasible on a small South African dataset spanning several languages, classifying calls by time-critical prehospital diagnosis from caller-described symptoms [8]; that work reported strong cross-validation accuracy that dropped once evaluated on held-out data, a pattern consistent with the generalisation gap this thesis investigates directly. Urgency classification from text alone has also been studied outside the emergency-call domain: comparing context-independent (TF-IDF) and context-dependent (BERT) representations for business voicemail triage [9], and, closer to this thesis's three-class formulation, using cross-lingual transformer encoders to classify multilingual medical-emergency social-media posts into three urgency tiers [10]. More recently, speech-act theory - the linguistic study of what a speaker is doing with an utterance, not just what it describes - has been applied to crisis-tweet urgency classification, showing that injecting speech-act information (assertions, orders, questions, subjective statements) into a transformer-based classifier measurably improves urgency and information-type detection, particularly when the model must generalise to an unseen type of disaster event [12]. That finding is directly relevant to this thesis's own generalisation question, even though it is applied to text-only social-media data rather than multimodal call audio. Closer still to this thesis's own design, a multimodal 9-1-1 call-prioritisation system combining acoustic features with speech-to-text transcription via late fusion has been proposed for English emergency calls, explicitly leaving multilingual, code-mixed, and non-English settings unaddressed [13]. In a similarly resource-constrained setting, an ASR-plus-NLU pipeline built from open-source models has been shown feasible for classifying unstructured emergency calls in a low- and middle-income country, Brazil, underscoring that off-the-shelf English-trained components do not transfer automatically to other languages [17]. Severity-prediction work closer to this thesis's own domain has also been pursued using one week of French fire-service call recordings, where a BERT-based classifier predicted serious-injury likelihood from caller speech alone at 71% accuracy [29], and using a deep ensemble multitask network combining clinical, demographic, and free-text dispatcher data to classify over a million Spanish emergency medical dispatch incidents by life-threatening

status, response-delay admissibility, and jurisdiction, outperforming an in-house triage protocol on all three tasks [26]. A simpler text-mining approach tuned specifically to raise recall for high-priority alerts, rather than overall accuracy, has also been proposed for a Latin American emergency-call system [32] - a prioritisation choice that mirrors this thesis's own emphasis on High-urgency recall over raw accuracy (Section 7.6).

Speech-based urgency and affect. Acoustic correlates of stress and arousal - fundamental frequency, jitter, energy dynamics, and speaking rate - are established markers in speech emotion recognition [4], with MFCCs the dominant paraverbal representation. Prior emergency-call classification work has generally relied on transcript content, leaving the acoustic channel comparatively underused despite its documented robustness to transcription failure - a property directly relevant to this thesis's findings on real, often acoustically noisy, telephony audio. Acoustic and prosodic features extracted from real telephone speech have also been shown to detect respiratory distress with high accuracy, reinforcing that paraverbal cues carry diagnostically useful information even over degraded telephony channels [18]. A decision-level, late-fusion approach combining acoustic and textual evidence for speech emotion recognition has likewise been shown to outperform unimodal baselines by allowing the more reliable modality to dominate on a per-sample basis [14], a finding consistent with this thesis's own late-fusion design (Section 3.5). Separately, real-time distress-detection work comparing CNNs against transformer-based acoustic models (Wav2Vec2, Whisper) found that transformer-based models generalise noticeably better to noisy and multilingual speech [15]. A broader systematic review of machine-learning approaches to speech emotion recognition confirms that acoustic-paraverbal evidence remains comparatively underexploited relative to text in applied, domain-specific settings such as this one [21]. Within an emergency-call setting specifically, attention-based fusion of speech and text has been explored directly on a French emergency call-centre corpus, finding that cross-attention fusion strategies outperform simple concatenation [23], and a separate study modelling caller emotion distribution in an emergency call centre found that redistributing calls by predicted emotional urgency could measurably reduce time-to-response for the most urgent callers [28].

Multilingual and code-mixed language. Cross-lingual encoders such as XLM-RoBERTa [1] transfer to low-resource languages with limited supervision, including specifically to urgency classification [10]. Weakly supervised multilingual speech recognition models [2] extend automatic speech recognition coverage across languages but are known, from the broader code-switching ASR literature, to degrade under intra-sentential code-mixing. Language identification for code-switched text has been established as a distinct modelling problem rather than noisy monolingual data [3], a position this component's feature design follows by treating Sinhala and Tamil context features separately. Separately, in the specific context of Sinhala natural language processing, recent work on Sinhala sentence embeddings demonstrates both the value and the difficulty of building effective NLP representations for a genuinely low-resource language: a two-tiered word-then-sentence embedding architecture was shown to outperform word embeddings alone on a Sinhala sentiment-analysis task, while the study also documents the practical obstacles - the absence of a mature Sinhala parser chief among them - that limit how far more sophisticated embedding techniques (such as hyperbolic embeddings) can currently be pushed for Sinhala [11]. This reinforces this thesis's decision to rely on rule-based, lexicon-driven textual/context features for Sinhala and Tamil rather than assuming a pre-trained Sinhala embedding model would be reliable for this domain. The difficulty of adapting ASR specifically to emergency-call speech in a low-resource language has been demonstrated directly for Azerbaijani, where domain-specific acoustic modelling and labelling choices materially affected recognition accuracy on real call-centre recordings [16]. For Sinhala specifically, a recent end-to-end speech-to-speech system combining ASR, retrieval-augmented response generation, and text-to-speech achieved strong transcription accuracy using a fine-tuned Wav2Vec2-BERT model, demonstrating that domain-adapted Sinhala ASR is achievable even with comparatively limited training data - though that system, unlike this component, was not built for the urgency-classification task or for code-mixed Sinhala-English input [20]. Closer to the specific Sinhala-English code-mixing this component must handle, dedicated NLP tooling has been built to process Sinhala text written in English letters (Singlish), a practical necessity given how Sri Lankans actually type on digital platforms [25]. For Tamil specifically, end-to-end speech-recognition work has

targeted the language directly [30], while a separate low-resource approach has shown that phoneme probability features from a pre-trained English ASR model can support Sinhala and Tamil speech-intent identification with as little as half an hour of in-language speech data per language [31]. At the level of full multilingual emergency-response NLU, a natural-language-understanding system for clinical triage across multilingual emergency dialogues has been proposed directly [27], and a preprint study has specifically targeted multilingual distress-call classification and detection with NLP techniques [33], though as an SSRN preprint it has not been through formal peer review.

Speaker diarization. Isolating a single speaker's audio from a multi-speaker recording - necessary here to separate the caller from the call-taker before acoustic feature extraction - is addressed by modern neural diarization pipelines such as pyannote.audio [7].

Class imbalance. The Synthetic Minority Over-sampling Technique (SMOTE) [6] is a standard method for correcting class imbalance during training without altering the evaluation distribution, used in this component inside cross-validation folds only, to avoid leakage.

Domain shift. That a model's held-out performance on its own training distribution does not predict its performance on a differently sourced dataset is a long-established finding in machine learning, sometimes termed dataset bias [5]. This thesis applies that concern directly and quantitatively, via Cohen's d, to the specific case of simulated-versus-authentic emergency audio, which, to the knowledge gathered during this review, has not previously been reported for Sri Lankan multilingual emergency calls specifically.

2.4 Existing systems products and techniques

Beyond academic research, operational emergency-call handling in most jurisdictions, including Sri Lanka, relies on human call-takers applying locally defined, often informal, triage criteria, supported at most by structured call-taking scripts or computer-aided dispatch (CAD) software that records and routes calls rather than classifying their urgency automatically. Commercial and research CAD-adjacent

systems exist for structured, scripted medical dispatch protocols in some countries, but these are generally built around standardised questioning scripts for specific emergency types, rather than a machine-learning classifier operating on arbitrary, multilingual, free-form caller speech. No existing system identified in this review combines automatic urgency classification, multimodal (acoustic and textual) evidence, and support for the specific multilingual, code-mixed context of Sri Lanka.

Table 2.1 compares this component against the closest research systems and techniques identified in this review, across the dimensions most relevant to its design.

Table 2.1: Comparison of this component against related research and techniques identified in the literature review.

System/study	Modality	Multilingual/code-mixed	Independent authentic-data evaluation	Urgency-specific
Anthony et al. [8]	Text (transcript)	Partial (multiple languages, not code-mixed)	Held-out split, same corpus	Yes
Trabelsi et al. [9]	Text	No	No	Yes (voicemail)
Singh et al. [10]	Text	Yes (cross-lingual)	No	Yes
Laurenti et al. [12]	Text (speech acts)	No (French only)	Out-of-event/out-of-type split, same corpus	Yes
Weeraprameshwara et al. [11]	Text (Sinhala embeddings)	Partial (Sinhala/English)	No	No (sentiment, not urgency)
Mirzaei et al. [13]	Acoustic + text (speech-to-text fusion)	No (English only)	No	Yes
Costa et al. [17]	Text (ASR transcript + NLU)	No (Portuguese only)	No	Yes
Ferri et al. [26]	Clinical + demographic + text (deep ensemble)	No (Spanish only)	No	No (severity/dispatch triage, not urgency tiers)
Abi Kanaan et al. [29]	Text (BERT, caller speech)	No (French only)	No	No (binary severity, not 3-tier urgency)
This component	Acoustic + text (multimodal)	Yes (Sinhala, Tamil, English, code-mixed)	Yes (independent DMC corpus)	Yes

2.5 Gap synthesis and conceptual framework

Taken together, the related research reviewed in Section 2.3 and the existing-systems comparison in Section 2.4 reveal a combined gap that no single prior study closes. First, urgency-classification research has generally reported performance on a single, internally consistent corpus - simulated or real - rather than developing a classifier on one and evaluating it independently on the other [8], [9], [10], [12]; this leaves open exactly the question that matters for deployment: does a promising development-set result actually transfer? Second, while the value of acoustic, paraverbal evidence is well established in speech emotion recognition [4], prior emergency-call urgency classification work has generally relied on transcript content alone, leaving the acoustic channel underused specifically in this application. Third, the multilingual, code-mixed character of Sri Lankan emergency communication means that findings and tooling built for monolingual or even general multilingual settings [1], [2], [3],

[11] cannot simply be assumed to hold; this component's own feature design treats Sinhala and Tamil context features separately for this reason. No study in this review combines these three elements, multimodal urgency classification, a multilingual and code-mixed Sri Lankan setting, and a quantitative, independently validated comparison of simulated-corpus against authentic-recording performance, for emergency-call triage.

This gap directly motivates the four research questions introduced in Section 1.6. RQ1 and RQ2 address the first and second points above by comparing unimodal and multimodal formulations. RQ3 and RQ4 address the first point directly, using the independent DMC evaluation and the domain-shift analysis to measure, rather than assume, how far simulated-corpus performance can be trusted. The conceptual framework carried forward into Chapter 3 is therefore a two-stage empirical design, development on a simulated corpus followed by independent validation on an authentic one, structured specifically to produce evidence on the transfer question the reviewed literature leaves open.

3 RESEARCH METHODOLOGY

3.1 Research design

This component follows an empirical, comparative machine-learning research design, carried out in two linked stages. The first stage is a development stage, in which unimodal and multimodal classifier formulations are built and compared under stratified five-fold cross-validation on a simulated development corpus, to answer RQ1 and RQ2. The second stage is an independent external-validation stage, in which every formulation from the development stage is evaluated, without retraining, against a separate corpus of authentic emergency-call recordings that played no part in development, to answer RQ3 and RQ4. The design deliberately treats the first stage's results as provisional: the final classifier formulation reported in this thesis is the one selected using evidence from the second, independent stage, not the one that scored highest during development.

This two-stage design was chosen specifically because a single-corpus evaluation - the pattern followed by most prior emergency-call classification research - cannot distinguish a classifier that has learned a genuinely transferable urgency signal from one that has learned patterns specific to how its development corpus happens to be constructed. Using two corpora collected through different processes, at different times, by different means, makes that distinction possible.

3.2 Data participants or study materials

Two corpora were used. The simulated development corpus consists of 267 role-played emergency calls, each labelled High (130 calls), Medium (82 calls), or Low (55 calls) urgency, constructed around fire and flood scenarios. A 40-call subset of this corpus was independently labelled by a second annotator to establish inter-annotator agreement; comparing the two sets of labels gives 32 exact matches out of 40 (80%) and a Cohen's kappa of 0.695 - “substantial” agreement on the standard Landis-Koch scale. Every disagreement in that subset involved the Medium class, which is consistent with Medium being the weakest-performing class throughout the later modelling results.

The independent validation corpus consists of 59 authentic recordings obtained from Sri Lanka's Disaster Management Centre (DMC), labelled High (23), Medium (14), or Low (22) urgency before any model was applied to them, and spanning nine districts. Unlike the simulated corpus, the DMC set is not scoped to a single incident type: flood accounts for less than half of the calls (25 of 59), with the remainder split across landslide, fallen-tree, and other weather-related incidents, a small number of fire calls, and general information requests, complaints, and enquiries. This corpus was collected entirely independently of the simulated corpus and is used both to evaluate generalisation and, directly, to select the final fusion formulation; it is described consistently throughout this thesis as an external validation set that informed model selection, not as an untouched final test set. Figure 3.1 compares the two corpora's class distributions directly.

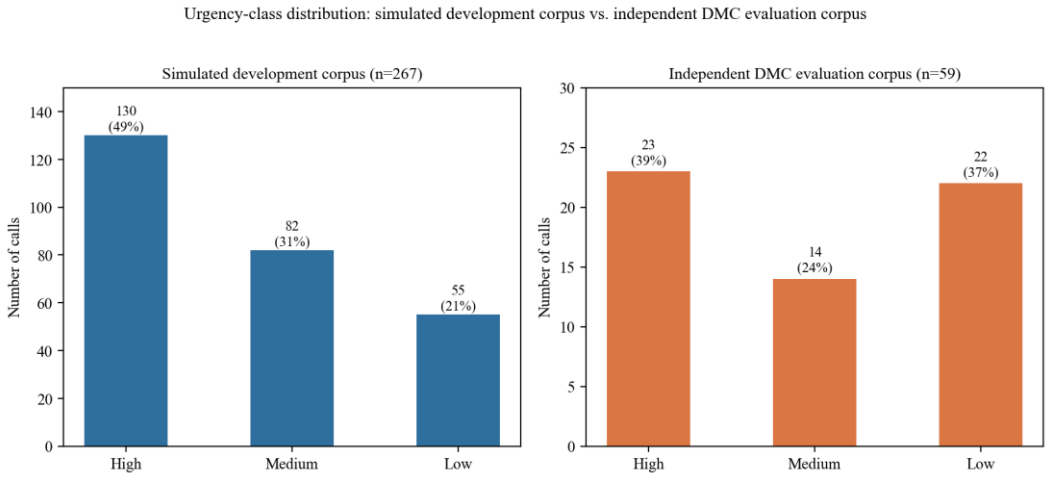


Figure 3.1: Urgency-class distribution, simulated development corpus vs. independent DMC evaluation corpus.

Both corpora were labelled using the same operational criteria, obtained directly from DMC staff in the absence of a formally documented urgency taxonomy: High for a call describing an immediate threat to life, Low for a purely informational or non-emergency call, and Medium for a present but non-immediate threat, where help is needed but not urgently. The absence of a precise, agency-defined boundary for the Medium class is itself part of why that class proves ambiguous, both in the inter-annotator agreement result above and in the modelling results reported in Chapter 6.

3.3 Experimental or study procedure

Audio preprocessing. Audio from both corpora is denoised, resampled to 16 kHz, and trimmed of leading and trailing silence before feature extraction, using the same pipeline for every recording. A preprocessing inconsistency identified during preliminary validation - which had been inflating the apparent cross-corpus shift of a single acoustic feature - was corrected before the experiments reported in this thesis.

Automatic speech recognition. Two ASR configurations were used, both routed by detected language (Sinhala versus Tamil/English) and differing only in the Sinhala model: a general-purpose Whisper-family model ("own-ASR"), used for the initial fusion-strategy comparison on the simulated corpus, and a domain-fine-tuned Sinhala model ("locked-ASR"), adopted after own-ASR was found to produce semantically incorrect Sinhala transcripts on real calls, used for the final external-validation configuration. This was a sequential adoption rather than a designed ASR ablation, so performance differences between the two configurations can reasonably be attributed to that substitution.

Feature extraction. Two feature families are extracted per call. The textual/conversational family comprises 19 features: an ASR-quality confidence score, keyword count, repetition rate, and sentiment polarity (three of which are weighted by ASR confidence), 14 language-specific rule-based context counts across Sinhala and Tamil (danger, help-request, high-risk-phrase, location-context, person-at-risk, safety-state, and a composite context score, each per language), and a shared fire/smoke keyword-match indicator. The keyword list behind the keyword-count feature was not chosen by hand alone: candidate terms were first surfaced through per-language TF-IDF contrasts between High, Medium, and Low calls, then curated manually. The acoustic-prosodic family comprises 91 features: 40 MFCCs summarised by mean and standard deviation, plus 11 prosodic and spectral descriptors (pitch mean and standard deviation, jitter, RMS-energy mean, standard deviation, slope, and gradient, zero-crossing rate, spectral centroid, speaking rate, and speaking-rate change).

Experimental conditions. Five evaluations were conducted, each mapped to a research question: (1) unimodal text-only and acoustic-only classification on the simulated corpus (RQ1); (2) early- versus late-fusion comparison under matched ASR (RQ2);

(3) an acoustic feature-subset ablation, to test whether the full 91-feature acoustic representation is necessary (RQ1, supporting evidence); (4) an independent evaluation of every formulation on the 59-call DMC set, used to assess generalisation and select the final configuration (RQ3); and (5) a quantitative domain-shift analysis comparing the simulated and DMC feature distributions, followed by a targeted audio-augmentation experiment testing the simplest candidate explanation for that shift (RQ4).

3.4 Measures and evaluation metrics

Classification performance is reported as macro-averaged F1 (macro-F1), which weights each of the three urgency classes equally regardless of how many calls fall into it, together with raw accuracy and per-class precision, recall, and F1. Macro-F1 is treated as the primary metric throughout, since raw accuracy alone can look artificially strong on an imbalanced three-class problem; High-urgency recall is reported alongside it because a classifier that always predicts High can achieve 100% High-recall mechanically, so recall on its own is an insufficient metric without macro-F1 for context.

On the independent DMC set, each model's prediction is its probability averaged across five training seeds, and uncertainty is reported as a 95% confidence interval obtained by non-parametric bootstrap resampling of the 59 DMC calls over 5,000 resamples. This reflects the uncertainty of evaluating on only 59 calls, rather than only the variance across training seeds. Inter-annotator agreement on the simulated corpus is measured using Cohen's kappa, and domain shift between the simulated and DMC feature distributions is measured per feature using Cohen's d, summarised as the median absolute value within each feature family.

3.5 Data analysis

Classification uses a Random Forest (200 trees, minimum leaf size 2, balanced class weighting) under stratified five-fold cross-validation on the simulated corpus, with feature standardisation and SMOTE class-balancing fitted inside each training fold only, to avoid leakage into validation folds. Simulated-corpus figures are averaged across five random seeds (1, 7, 42, 99, 123) for stability. Feature selection -

specifically, removing a context feature that depressed Medium-class recall and a lexical feature that proved to be a near-noise, heavily domain-shifted signal, and adding a fire/smoke keyword-match indicator - was carried out by comparing configurations across the full development corpus ahead of the cross-validation reported in this thesis, which is distinct from within-fold selection; the reported cross-validation estimate is therefore not strictly leakage-free in the within-fold sense, and is reported as such throughout rather than presented as an unqualified unbiased estimate. That selection process was targeted and evidence-driven for the specific features where there was a documented reason to suspect them, together with block-level ablation of the acoustic feature family, rather than an exhaustive independent test of every one of the 91 acoustic and 19-20 textual/conversational features individually. Table A.4 reports the full progression of this feature-selection ablation, ending in the frozen 111-feature configuration used throughout this thesis.

For the final late-fusion formulation, two Random Forests - one trained on each feature family - are trained independently on the simulated corpus and combined only at inference time, by averaging their predicted class probabilities. This formulation, and its fusion weighting, was selected using its behaviour on the independent DMC evaluation, not on simulated-corpus cross-validation performance alone.

3.6 Ethics privacy security and responsible AI

Ethical approval and data handling. Use of authentic emergency call recordings from the Disaster Management Centre required explicit coordination with DMC regarding data-sharing terms and the ethical handling of recordings that may contain identifying or distressing content. Recordings and transcripts were handled and stored under access controls limited to the research team, and are not published alongside this thesis; only aggregated, de-identified statistics and short illustrative transcript fragments, where needed, are reported. The simulated corpus, being role-played by consenting participants rather than drawn from real emergencies, carries lower privacy risk but was still collected and stored under the same access-control practices.

Responsible AI considerations. This component is explicitly scoped as a decision-support signal, not an autonomous dispatch or safety-critical system: no claim is made

anywhere in this thesis that the classifier offers a clinical or safety guarantee. Three responsible-AI risks are addressed directly in the methodology and results rather than left implicit. First, class imbalance and label ambiguity (the Medium class in particular) are measured and reported rather than smoothed over, since a model that is confidently wrong about a borderline case is a worse outcome than one that is visibly uncertain. Second, the independent DMC evaluation exists specifically to check whether simulated-corpus performance is being over-trusted; the results in Chapter 6 show that it is, for several formulations, and that finding is reported prominently rather than minimised. Third, High-urgency recall is tracked as a distinct, operationally important metric, since under-flagging a genuinely high-urgency call is judged a more costly error than over-flagging a lower-urgency one, even though this thesis does not formally define a cost matrix establishing that hierarchy.

AI tool use disclosure. Generative AI tools were used during this research and in the preparation of this thesis, in accordance with the module's academic integrity policy. The nature, scope, and verification of that use is disclosed in full in Appendix B; the candidate remains responsible for the accuracy, originality, and correctness of all content, code, data, analyses, and conclusions presented.

3.7 Reproducibility and research artefacts

The component's codebase is organised as a pipeline (feature extraction, training, and DMC evaluation scripts) together with a numbered series of analysis folders covering error diagnosis, feature ablation, fusion experiments, domain-shift and scope analysis, ruled-out negative results, final-candidate tuning, and general audits and utilities. Each experimental script states its purpose, inputs, and outputs at the top of the file, and disposable diagnostic scripts are kept separate from the frozen production pipeline.

The final feature set and classifier configuration are recorded in a dedicated pipeline-freeze document that fixes the exact feature list, model hyperparameters, and random seeds used for the reported results, so that the reported figures can be regenerated from the same inputs. Simulated-corpus results are averaged over five fixed random seeds, and DMC bootstrap confidence intervals use a fixed number of resamples, for the same reason. Several negative results - configurations that were tried and did not improve

on the reported formulation, such as hyperparameter re-tuning, probability calibration, and ASR-hallucination-detection heuristics - are kept in the repository and summarised in Chapter 7 rather than discarded, so that the final methodology's reasoning remains traceable. The authentic DMC recordings themselves are not distributed with this thesis for privacy reasons; the simulated corpus and all derived, de-identified feature tables are available on request, subject to the data-handling terms agreed for this project.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements for this component are: (FR1) ingest a call audio file and detect its language(s); (FR2) extract the defined acoustic-prosodic and textual/conversational feature sets from the audio and its transcript; (FR3) produce an urgency label (High, Medium, or Low) together with the underlying class probabilities from each modality-specific model; (FR4) support Sinhala, Tamil, English, and code-mixed combinations of these; and (FR5) expose its output in a structured, machine-readable format suitable for consumption by the group's multi-source incident-fusion component.

Non-functional requirements are: classification quality, measured primarily by macro-F1 and High-urgency recall rather than raw accuracy, given the class imbalance in both corpora (Section 3.4); robustness to unreliable or partially incorrect transcripts, addressed through ASR-confidence weighting (Section 3.3) and the acoustic channel's independence from transcript quality; reproducibility, through a version-controlled, deterministic feature-extraction and classification pipeline (Section 3.7); and privacy and data-handling compliance for authentic call recordings (Section 3.6). Formal latency and throughput requirements have not yet been defined or benchmarked for this component, which is identified as outstanding work in Section 6.2 and Chapter 9. Table 4.1 traces each requirement to where it is addressed and verified in this thesis.

Table 4.1: Requirements traceability.

Requirement	Description	Addressed in	Verified in
FR1	Ingest call audio and detect language(s)	Section 4.2	Section 5.2; Section 6.1 (all 59 DMC calls processed)
FR2	Extract acoustic-prosodic and textual/conversational features	Section 3.3, Section 4.4	Section 5.4 (extraction-path verification script)
FR3	Produce an urgency label with per-modality class probabilities	Section 4.2, Section 4.4	Appendix A (worked example); Section 5.3 (output format)

Requirement	Description	Addressed in	Verified in
FR4	Support Sinhala, Tamil, English, and code-mixed input	Section 3.3, Section 4.3	Section 6.1 (DMC spans all three languages)
FR5	Expose output in a structured format for incident fusion	Section 5.3	Partially verified; end-to-end integration test outstanding (Section 6.4)
NFR - Classification quality	Macro-F1 and High-urgency recall as primary metrics	Section 3.4	Section 6.1 (Figure 6.1, Figure 6.2)
NFR - Robustness to ASR errors	ASR-confidence weighting; acoustic channel independence	Section 3.3	Section 6.1 error analysis (ASR_FAILURE category)
NFR - Reproducibility	Version-controlled, deterministic pipeline	Section 3.7	Section 5.4 (determinism check); Appendix A (reproduction commands)
NFR - Privacy/data-handling	Restricted access, de-identified reporting	Section 3.6	Not independently audited beyond DMC agreement
NFR - Latency/throughput	Not yet defined	Not addressed	Outstanding (Section 6.2, Section 9.4)

4.2 Component architecture within the complete solution

This component sits at the start of the group's four-component pipeline, as the stage that turns a raw call recording into a structured urgency signal. Internally, it is organised as a sequential pipeline: audio preprocessing (denoising, resampling to 16 kHz, silence trimming), speaker diarization to isolate the caller's audio from the call-taker's, language-routed automatic speech recognition, parallel extraction of the acoustic-prosodic and textual/conversational feature sets, classification by the locked late-fusion model (an acoustic-only Random Forest and a text-only Random Forest combined by averaging their predicted class probabilities), and finally packaging of the predicted label and probabilities into the output format consumed downstream.

Within the complete four-component system, this urgency-classification output is intended to be one input among several into the multi-source incident-fusion

component (Component 4), alongside structured information produced by the AI agent and SMS intelligence components. This component does not itself make a dispatch decision; its architectural role is strictly to produce a well-calibrated, evidence-based urgency signal that the fusion layer and, ultimately, human dispatch staff can act on. Figure 4.1 shows the full pipeline, from raw call audio through the two independently-trained branches to the fused urgency label.

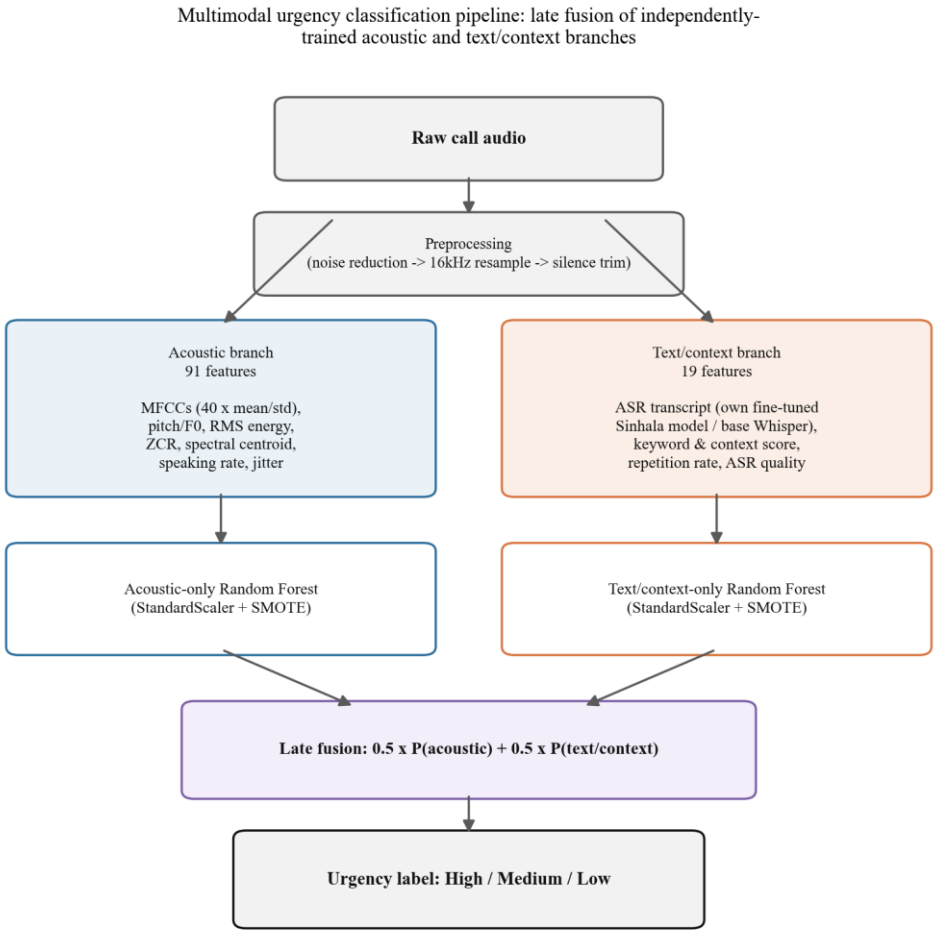


Figure 4.1: Multimodal urgency-classification pipeline — late fusion of independently-trained acoustic and text/context branches.

4.3 Design alternatives and decisions

Several design alternatives were evaluated and explicitly rejected during development, each on the basis of evidence rather than convention. Early fusion - concatenating both

feature families into a single model - was the strongest formulation on the simulated development corpus, and was the initial design choice; it was replaced by late fusion only after the independent DMC evaluation showed it did not generalise (Section 6.1, Section 7.1). The own-ASR configuration was the initial choice for automatic speech recognition, and was replaced by a domain-fine-tuned Sinhala model (locked-ASR) after own-ASR was found to produce semantically incorrect Sinhala transcripts on real calls. Within feature design, word_count and a per-language immediacy-count feature were both evaluated and dropped - the former for being a near-noise, heavily domain-shifted signal, the latter for depressing Medium-class recall - while a fire/smoke keyword-match indicator was evaluated and retained after it improved development-set performance (Section 3.5).

Beyond the feature and fusion decisions, several further alternatives were tested specifically to try to close the simulated-to-authentic performance gap, and were not adopted because they did not provide a sufficient, justified improvement. Hyperparameter re-tuning of the Random Forest classifiers gained 0.6 points of macro-F1 on simulated cross-validation but reduced scope-matched DMC accuracy from 67.3% to 63.6%, indicating the simulated-optimal parameters did not transfer; probability calibration (Platt/sigmoid scaling) worsened the Brier score across all 5 tested seeds, consistent with there being too little simulated-corpus data (267 calls) for reliable calibration; and a heuristic compression-ratio detector for ASR hallucinations dropped DMC macro-F1 from 0.449 to 0.285 and did not recover High-urgency classification on any of the specific misclassified calls it targeted. These are recorded as ruled-out negative results rather than discarded, so that the reasoning behind the final design remains traceable (Section 3.7).

4.4 Detailed design

The classifier itself is a Random Forest (200 trees, minimum leaf size 2, balanced class weighting), chosen for its robustness on a small, imbalanced, tabular dataset and its resistance to overfitting relative to more complex models given the dataset sizes involved. Feature standardisation and SMOTE class-balancing are fitted inside each training fold only, to prevent information from validation data leaking into training. Simulated-corpus results are averaged across five fixed random seeds for stability, and

the final DMC prediction for each model is its five-seed-averaged class probability, with uncertainty reported as a 95% confidence interval from 5,000-resample non-parametric bootstrap resampling of the 59 DMC calls. This choice was tested directly rather than assumed: on the same simulated 5-fold cross-validation, a logistic-regression stacking meta-learner over the text-only and acoustic-only out-of-fold probabilities reached 64.9% accuracy and a Gradient Boosting classifier on the fused feature set reached 63.4%, both below the Random Forest baseline's 65.7%; neither alternative was pursued further on that basis (scripts/analysis/06_final_candidate_and_tuning/push_to_70_simulated.py).

The ASR-confidence weighting mechanism multiplies three of the lexical textual features (keyword_count, repetition_rate, sentiment_polarity) directly by a per-utterance ASR-quality score, so that a low-confidence transcript automatically contributes less to those specific features rather than being taken at face value; the 14 language-specific context counts and the fire/smoke indicator are computed directly from the transcript without this weighting. The late-fusion combination itself is a simple, interpretable unweighted average of the two modality-specific models' predicted class probabilities, selected over several more complex weighting schemes (a cross-validation-tuned acoustic weight, per-class weighting, and confidence-gated fusion) because it performed best on the independent DMC evaluation that determined the final design (Section 3.5).

4.5 Standards and professional practice

This component follows several professional-practice conventions intended to make its results trustworthy and its design auditable. All citations and the reference list follow the IEEE citation style (Section references, Chapter 2). Feature selection, model configuration, and random seeds are fixed and recorded in a version-controlled pipeline-freeze document, rather than left to vary between runs, so that reported results can be regenerated. Negative results - configurations that were tried and did not improve on the adopted design - are retained and documented rather than silently discarded (Section 3.7, Section 4.3), which follows the broader professional norm that a system's design history should be traceable, not just its final state.

Handling of the authentic DMC recordings follows the data-protection and consent practices agreed with the Disaster Management Centre and described in Section 3.6, including restricted access and de-identified reporting.module's academic integrity policy, the use of generative AI tools in this research and in the preparation of this thesis is disclosed in full in Appendix B, consistent with the responsible-AI practice of being transparent about how a system, and the document describing it, were actually produced.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The component is implemented in Python, using scikit-learn for the Random Forest classifiers, feature standardisation, and stratified cross-validation; imbalanced-learn for SMOTE class-balancing; librosa for acoustic feature extraction (MFCCs and the prosodic/spectral descriptors); pyannote.audio for speaker diarization; and Whisper-family models (a general-purpose base model for own-ASR, and a domain-fine-tuned Sinhala model for locked-ASR) for automatic speech recognition. pandas and numpy are used throughout for data handling, and matplotlib is used for generating diagnostic and reporting plots, including the confusion-matrix figure presented in Chapter 6. librosa was chosen over the domain-standard openSMILE toolkit [24] for acoustic feature extraction primarily for its tighter Python integration with the rest of the pipeline, not because of any measured accuracy difference between the two.

Development was carried out locally, with experiments organised as standalone, independently runnable Python scripts rather than a single monolithic application, to keep each experiment's purpose, inputs, and outputs traceable (Section 3.7). Version control was used throughout to track changes to the production pipeline, separate from the larger set of disposable diagnostic and ablation scripts described below.

5.2 Component implementation

The production pipeline is organised into clearly separated stages: audio preprocessing and feature extraction scripts (covering acoustic feature extraction, caller-turn extraction from diarized audio, and textual/context feature extraction from transcripts), a training script that fits the locked classifier configuration on the simulated corpus, and a family of DMC evaluation scripts that apply the frozen pipeline to the independent authentic recordings. Several iterations of the DMC evaluation script exist (reflecting the ASR configuration change from own-ASR to locked-ASR and subsequent refinements), with the final, reported configuration explicitly identified in the pipeline-freeze document rather than left ambiguous between versions.

Alongside the production pipeline, a substantially larger set of analysis scripts was implemented to support the experimental work described in Chapter 3 and Chapter 6:

feature-ablation experiments (testing acoustic feature subsets and individual textual features), fusion-strategy experiments (comparing early and late fusion and several fusion-weighting schemes), domain-shift and scope-matching diagnostics, error-diagnosis scripts for understanding specific misclassifications, and a set of scripts recording the ruled-out negative results discussed in Section 4.3. Each experimental script states its purpose, inputs, and outputs at the top of the file, and is kept separate from the frozen production pipeline so that disposable, exploratory work cannot accidentally alter the reported, locked configuration.

The integration surface itself - the module the other components actually call - is urgency_bridge.py, which exposes two entry points sharing the same contract: classify_urgency(), which runs the single 111-feature joint model, and classify_urgency_late_fusion(), which runs the locked late-fusion configuration by combining a text-only and an acoustic-only model's predicted probabilities at inference time. Both accept a per-turn audio clip, its transcript, a detected language code, and an optional session identifier, and return a label, a confidence score, and the full class-probability distribution; no automatic speech recognition happens inside this module, since the calling component (the AI agent) has already transcribed the turn, and re-transcribing it here would add unnecessary latency to a live call.

Two implementation details were deliberately built to handle real, messy call audio rather than only the clean inputs the development corpus provides. First, a turn shorter than one second, or one that contains nothing but silence after denoising and trimming, is not forced through the classifier; it is returned as an explicitly skipped result with a stated reason, so that a near-empty turn cannot silently produce a misleadingly confident prediction. Second, when a session identifier is supplied, each turn's result is appended to a per-call session log, and a separate aggregation function, get_call_urgency(), combines every non-skipped turn in that call into a single call-level verdict by averaging the turns' class-probability vectors and taking the argmax - a rule fixed before any result was seen, to avoid post-hoc-tuning the aggregation itself. The single highest-severity turn in the call is also reported alongside the averaged verdict, rather than being discarded or overwritten by it, on the reasoning that one clearly High-urgency turn is operationally significant even in a call that averages

calmer overall; end_call() waits for any turns still being processed asynchronously before computing this final aggregation, so a verdict is never produced from an incomplete set of turns.

A standalone Flask web demo (app.py) was also built to exercise this exact pipeline independently of the live call-handling integration: a user uploads or records an audio file through a browser, and the application runs it through the same preprocessing, feature extraction, and locked late-fusion classifier as the integrated system, calling urgency_bridge.py directly rather than a separate reimplementation, so that a result produced through the demo is guaranteed to match what the integrated system would have produced for the same audio. This served as the primary tool for manually auditing individual predictions during development and for the DMC Director General demonstration described in Section 6.3.

5.3 Integration with the other components

This component's intended integration point with the rest of the group's system is a structured output record - an urgency label and the associated class probabilities from each modality-specific model - produced per call, for consumption by the multi-source incident-fusion component (Component 4). The output format was designed to be modality-transparent, exposing the acoustic-only and text-only probabilities separately rather than only the final fused label, so that the fusion component (or a human reviewer) can see which modality is driving a given prediction, which is particularly relevant given this thesis's finding that the two modalities do not always agree and do not transfer equally well to authentic recordings (Chapter 6, Chapter 7).

As reported honestly in Section 6.4, end-to-end integration testing in which this component's output is actually consumed by the incident-fusion layer alongside the AI agent and SMS intelligence components had not been completed at the time of writing this thesis. This reflects the current stage of the group project as a whole rather than a limitation specific to this component, and completing that integration is identified as a priority item in Section 9.4.

5.4 Verification and software testing

Verification of this component relies on several complementary checks rather than a single test suite. A dedicated extraction-path verification script regenerates acoustic and textual features for calls already present in the feature table, starting from the same audio, and checks that the regenerated values are identical to the stored ones; this proves that the feature-extraction path itself is correct and deterministic, independent of any modelling choices. Structural assertions are used throughout the codebase to check that each named feature-set variant has exactly the expected number of columns (for example, confirming the acoustic set is exactly 91 features and the frozen full feature set is exactly 111), catching accidental feature-list drift before it can silently affect results.

Model-level verification relies on statistical rather than unit-test methods, appropriate to a machine-learning component: five-seed averaging on the simulated corpus to check that results are not an artefact of a single random initialisation, and 5,000-resample non-parametric bootstrap confidence intervals on the independent DMC evaluation to check that reported differences between formulations are not within noise. The ruled-out negative-results scripts (Section 4.3) also serve a verification purpose, in that they document configurations that were tested and did not hold up, reducing the risk of the final design resting on an untested assumption. Formal software testing in the conventional sense - a unit or integration test suite with continuous-integration enforcement - has not been built for this component and is identified as future work in Section 9.4.

5.5 Deployment and operation

At the time of writing, this component operates as a research-grade, script-based pipeline rather than a deployed, always-on service: feature extraction, training, and DMC evaluation are run as discrete batch jobs, not behind an API. This is appropriate to the component's current stage of validation - given the findings in Chapter 6 and Chapter 7, particularly around the simulated-to-authentic performance gap and the as-yet-untested four-component integration, packaging the component as a production service ahead of that validation being complete would risk operationalising a configuration before its real-world behaviour is sufficiently understood.

Moving this component into a deployed, operational state - for example, as a service the incident-fusion component can call per incoming audio file, with defined latency and throughput characteristics - is identified as a concrete next step once the outstanding validation work described in Sections 6.2 to 6.4 is complete, and is discussed further as future work in Section 9.4.

6 RESULTS AND VALIDATION

6.1 Research results

On the simulated development corpus, under stratified five-fold cross-validation, the 91-feature acoustic-only model reached macro-F1 0.618 and the 19-feature text-only model reached macro-F1 0.397. Combining both families into a single 111-feature early-fusion model raised this to macro-F1 0.630, the best matched-ASR development-set result obtained, ahead of late fusion's 0.613 under the same conditions. These figures follow feature selection performed on the same corpus ahead of the reported cross-validation (Section 3.5), and are reported with that caveat throughout rather than treated as a fully leakage-free estimate. Figure 6.1 compares all four formulations on both the simulated corpus and the independent DMC set.

The independent evaluation against the 59-call DMC corpus is the central result of this thesis. Every formulation's DMC performance sits below its simulated-corpus estimate to some degree, with one notable exception: text-only macro-F1 actually rose from 0.397 on the simulated corpus to 0.449 on DMC, while early fusion - the strongest model during development - fell to 0.326 macro-F1 on DMC (95% CI 0.234-0.470), close to the majority-class baseline of 0.187. This reversal is only visible because of the independent evaluation, and it directly answers RQ3: a classifier selected purely on simulated-corpus performance would have been the wrong choice for authentic recordings. The final late-fusion formulation (acoustic-only and text-only models combined via unweighted probability averaging, under the locked-ASR configuration) reached 43.7% accuracy, 0.428 macro-F1, and, its clearest strength, 46.1% High-urgency recall on DMC Full - the strongest High-urgency recall among the learned classifier configurations tested, though text-only alone remained comparable or better on raw DMC accuracy. Figure 6.2 and Figure 6.3 show the final model’s per-class recall and full confusion matrix on DMC Full. On the 11-call fire/flood-scope-matched subset - the DMC calls whose incident type matches what the simulated training corpus was actually built around - the same late-fusion model reaches 67.3% accuracy, 0.630 macro-F1, and 65.7% High-urgency recall, materially higher than on DMC Full; given the subset is just 11 calls, this gap is read as directional evidence that part of the DMC-

Full gap reflects scope mismatch rather than model failure (Section 7.5), not as a precise estimate.

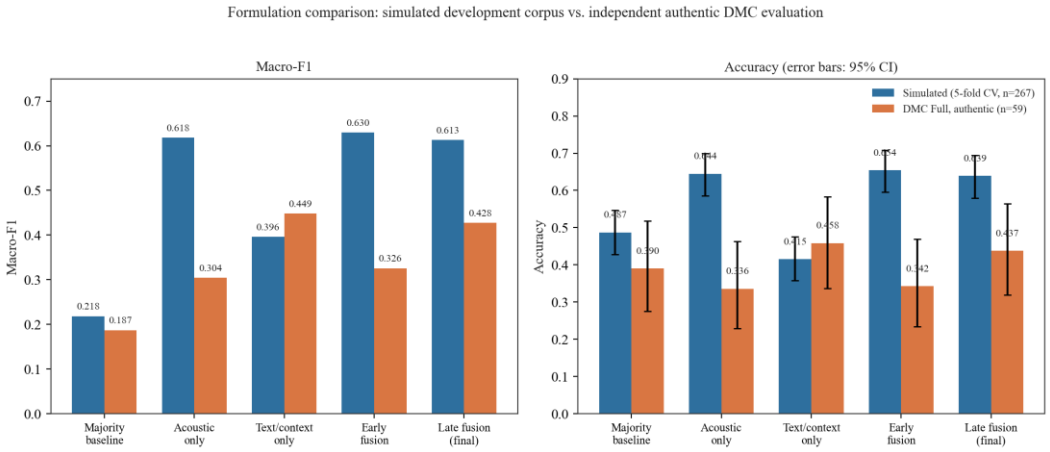


Figure 6.1: Macro-F1 and accuracy by formulation, simulated development corpus vs. independent DMC evaluation (95% CI on accuracy).

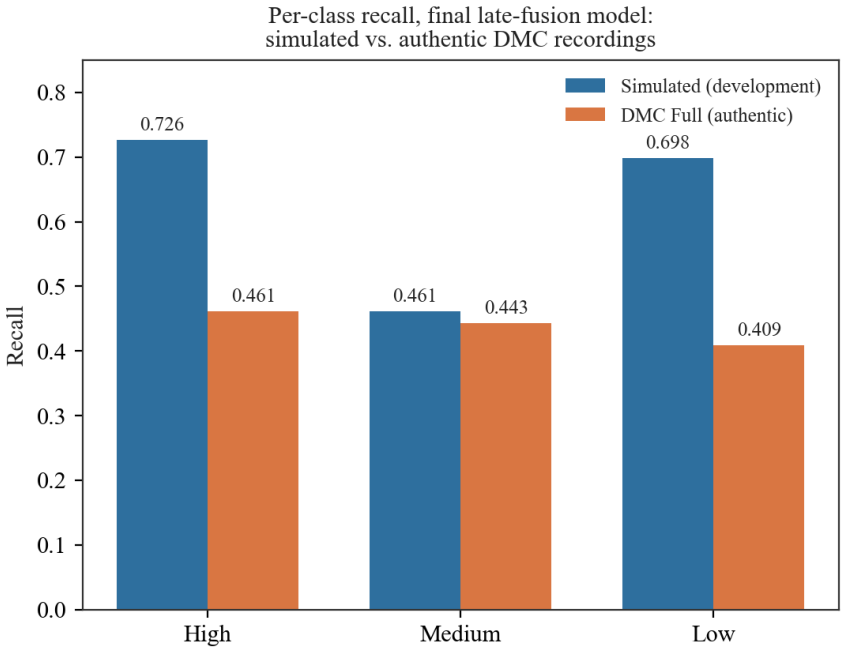


Figure 6.2: Per-class recall of the final late-fusion model, simulated vs. authentic DMC recordings.

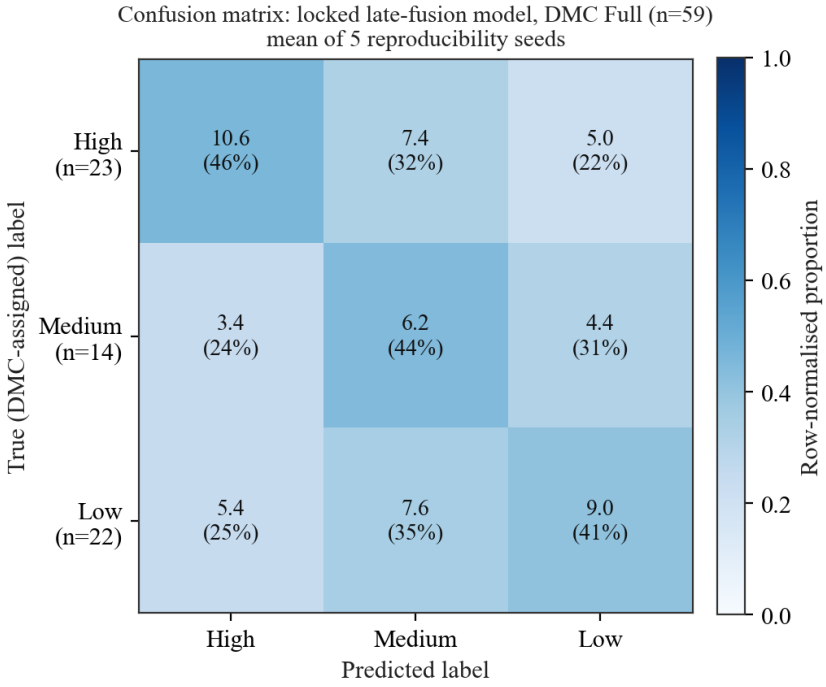


Figure 6.3: Confusion matrix of the final late-fusion model on DMC Full (n=59), mean of 5 reproducibility seeds; full precision/recall/F1 per class in Appendix A.

The domain-shift analysis addressing RQ4 found that the acoustic feature family shifted considerably more between the two corpora (median |Cohen's d| = 0.57) than the text/context family (median |d| = 0.154), with 25 of 91 acoustic features showing a large shift (|d| > 0.8) against essentially none of the text/context features. A targeted audio-augmentation experiment - adding white noise and a telephone-bandwidth bandpass filter to the simulated corpus - increased, rather than reduced, the measured domain shift, indicating that simple telephony-channel differences do not by themselves explain the acoustic mismatch. Figure 6.4 shows the individual features with the largest shift, and Figure 6.5 visualises the simulated and DMC acoustic feature distributions directly via PCA.

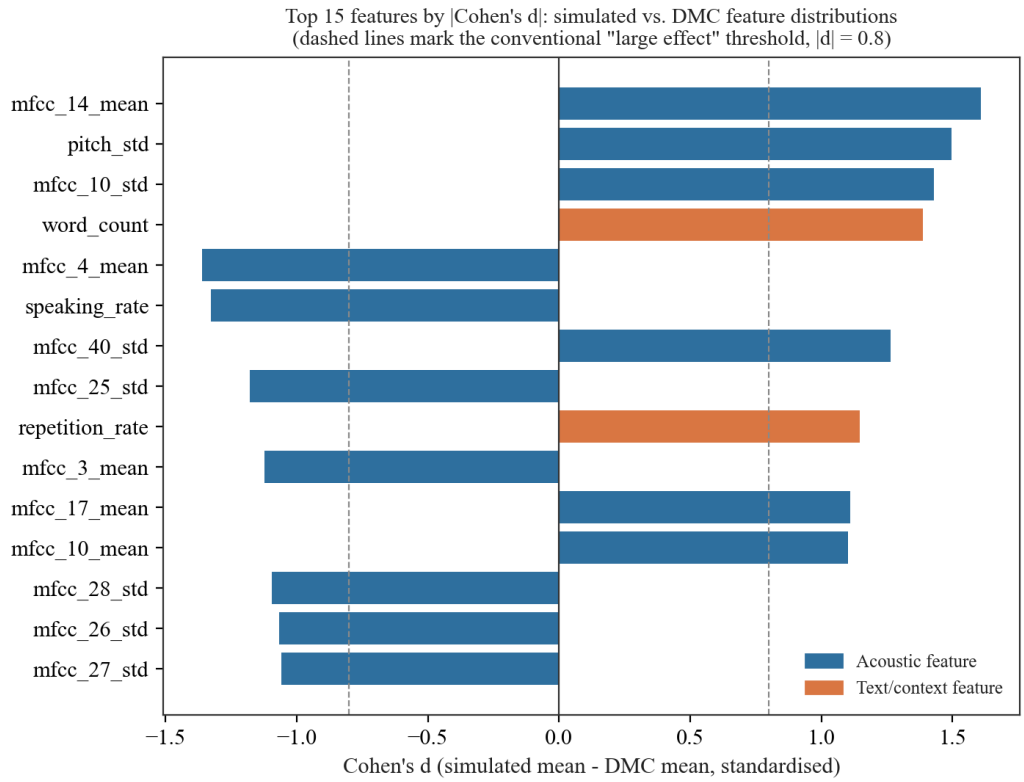


Figure 6.4: Top 15 features by |Cohen's d| between the simulated and DMC feature distributions.

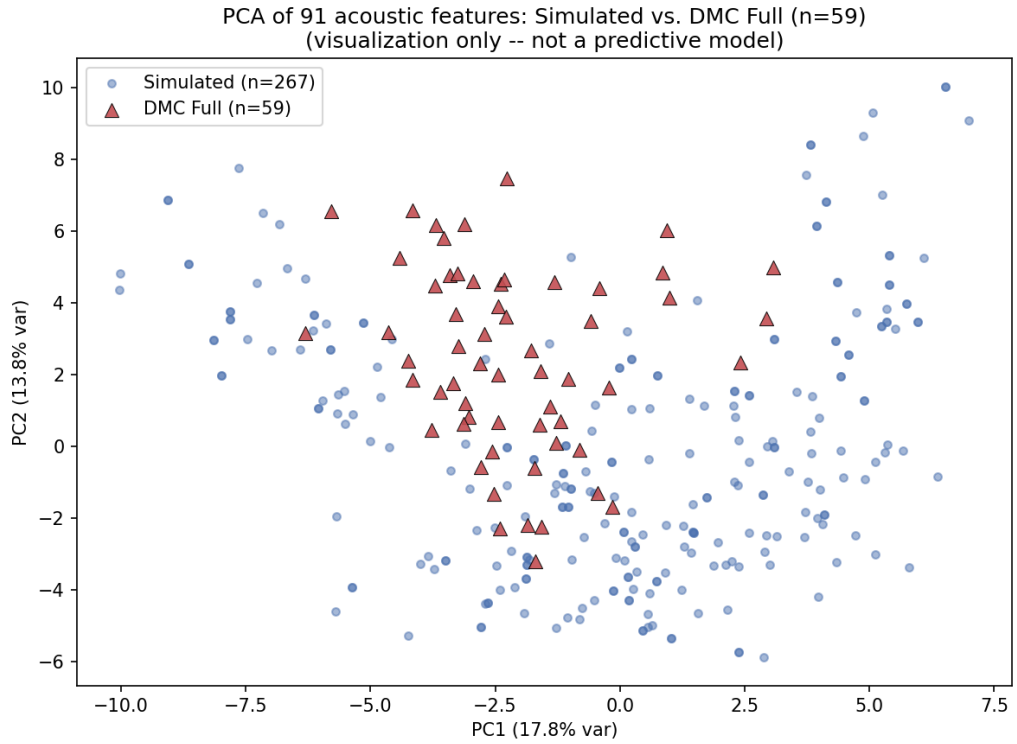


Figure 6.5: PCA projection of the 91 acoustic features, simulated corpus vs. DMC Full (visualisation only, not a predictive model).

Error analysis. To understand not just how much performance degraded but why, every one of the early-fusion model's 59 DMC predictions was manually categorised by root cause. Of the 39 errors (59 calls, 33.9% correct), the single largest category was transcription failure: 20 errors (34% of all 59 calls) were cases where the Sinhala/Tamil keyword-count and context-score features were all zero because no recognisable context words appeared in the transcript at all - the text branch had essentially nothing to work with, independent of how the model weighted it. A further 10 errors (17% of calls) were cases where the retained class-probability margins show the acoustic evidence was actually closer to the true class than to the model's prediction, yet the single joint model chose the wrong class anyway - direct evidence that a model trained on concatenated features can let one modality's noise override a cleaner signal in the other, the specific risk identified as a candidate explanation for early fusion's DMC underperformance in Section 7.1.

The equivalent root-cause breakdown of the 98 misclassifications from the simulated-corpus cross-validation (the same early-fusion configuration) shows a consistent pattern: 46 of 98 errors (47%) were cases of valid transcription with no keyword or context signal present, and a further 15 involved transcripts that were confidently but incorrectly recognised in the wrong script or language. Errors also concentrated specifically around the Medium class: 33 of 98 simulated-corpus errors were Medium calls predicted as High, and a further 13 were Medium predicted as Low, together accounting for nearly half of all misclassifications - consistent with the Medium class's

comparatively weak inter-annotator agreement (Section 3.2) and its lack of a precise, agency-defined boundary (Section 3.2).

6.2 Product and technical results

From a product perspective, the component operates as a batch or per-call classification service: given a call audio file and detected language, it produces an urgency label (High, Medium, or Low) together with the underlying class probabilities from each modality-specific model. The ASR-confidence weighting mechanism (Section 3.3) means the component degrades gracefully rather than failing outright when transcription quality is poor on a given call, since low-confidence transcripts automatically contribute less to the affected textual features rather than being trusted at face value. The acoustic feature pipeline does not depend on transcript quality at all, which is its principal technical value as a fallback signal when ASR output is unreliable - a condition that occurs routinely on real DMC audio given the code-mixed, colloquial nature of the speech.

The locked feature-extraction and classification pipeline is deterministic and fully specified in a version-controlled configuration (Section 3.7), so that a given audio file and transcript combination always produces the same prediction. Formal latency and throughput benchmarking of the end-to-end pipeline (audio preprocessing, diarization, ASR, feature extraction, and classification) on production-representative hardware has not yet been carried out as part of this thesis and is identified as outstanding work in Chapter 9.

6.3 User or stakeholder validation

Formal user or stakeholder validation - for example, a structured usability study with DMC call-takers or dispatch staff - was not conducted as part of this thesis. The closest validation activity undertaken was informal: consultation with DMC staff, including a discussion with the DMC Director General during the same period in which authentic call recordings were being obtained from DMC, to establish the operational criteria used to label both the simulated and DMC corpora (Section 3.2). This grounded the urgency label definitions in how DMC itself distinguishes High, Medium, and Low cases, rather than in an arbitrary academic definition. In that discussion, after the approach and system were explained, the Director General indicated that a system of

this kind would be useful to DMC's operations, and requested that the completed system be presented to DMC once the research concludes. That presentation is planned as a direct follow-on from this thesis rather than something that has taken place yet. This positive reaction is acknowledged as exactly what it is - one senior stakeholder's informal, individual response - and not a structured usability study: the classifier's output has not been shown to a call-taker in a live or simulated dispatch setting, so its usability and trustworthiness in actual operational use remain untested. Completing the planned presentation to DMC, and conducting a structured usability evaluation with call-taking staff specifically, are identified as priority future work in Chapter 9.

6.4 Integration results

The component is designed to expose its prediction - an urgency label and the associated class probabilities - in a structured, machine-readable format intended for consumption by the multi-source incident-fusion component described in Chapter 1. End-to-end integration testing across all four components of the group project, in which a call processed by this component's output is actually consumed by the fusion layer alongside output from the AI agent and SMS intelligence components, had not been completed at the time of writing this thesis, and is scheduled to be carried out the following week. This reflects the current stage of the group project rather than a limitation of this component specifically, and is reported honestly here as planned work rather than presented as already completed.

6.5 Summary of findings

In summary: acoustic and textual features each carry a usable urgency signal individually (RQ1), and multimodal fusion improves on acoustic-only during development, though the fusion strategy matters a great deal once authentic data is introduced (RQ2). Simulated-corpus performance does not reliably predict authentic-recording performance, and the direction of that gap is modality-dependent rather than uniform (RQ3). Acoustic features shift substantially more than textual features between the two corpora, which helps explain, though does not fully account for, why acoustic-heavy formulations transfer less effectively (RQ4). The classifier's technical behaviour is well characterised by these results; its behaviour as a deployed product

component, validated against real users and integrated with the rest of the group's system, remains the main outstanding work going into the final weeks of this project.

7 DISCUSSION

7.1 Interpretation of findings

RQ1 - individual modalities. Acoustic features alone reach 0.618 macro-F1 on the simulated corpus but transfer poorly to DMC (0.304 macro-F1, close to the majority baseline). Text/conversational features are a noticeably weaker predictor of urgency on the simulated corpus by themselves, yet become the strongest single modality on the independent DMC set by raw accuracy - a reversal that is only visible because of the external evaluation, and one of the clearest findings of this thesis.

RQ2 - fusion strategy. Early fusion's simulated-corpus lead over acoustic-only (0.630 vs. 0.618) disappears entirely on DMC, where it ends up the lowest-scoring formulation on every metric. One possible explanation is that a single tree ensemble spanning both modalities picks up on acoustic/lexical co-occurrence patterns specific to how the simulated corpus was produced, patterns that training the two branches independently would limit; this thesis's design does not isolate that mechanism directly, and it is identified as a candidate explanation rather than a proven one. Late fusion, despite not being the top scorer on simulated-corpus cross-validation, was the formulation whose DMC-evaluation behaviour justified selecting it as the final choice. The error analysis in Section 6.1 gives partial, direct support for this: in 10 of the 39 DMC errors, the retained acoustic evidence alone pointed to the correct class while the single joint model predicted the wrong one anyway, consistent with one modality's noise overriding a cleaner signal in the other inside a shared model - though this still does not isolate the mechanism as cleanly as a dedicated ablation would.

RQ3 - simulated-to-authentic generalisation. Most multimodal and acoustic formulations show a substantial gap between simulated-corpus and DMC performance, while text-only is an exception, achieving higher DMC macro-F1 than its simulated-corpus estimate; that modality-dependent gap is this thesis's central finding. Of the formulations tested, the one selected purely on development-set macro-F1 (early fusion, 0.630) did no better on DMC than the acoustic-only model it had clearly beaten in development.

RQ4 - explaining the gap. The domain-shift analysis shows the acoustic feature space shifted considerably more than the text feature space between the two corpora, giving a partial, feature-level explanation for why acoustic-heavy formulations transferred less effectively than text-heavy and late-fusion formulations. This relationship is an association consistent with the observed transfer results, not evidence of causation.

7.2 Comparison with literature baselines and existing solutions

Compared against the research reviewed in Chapter 2 (summarised in Table 2.1), this thesis's approach differs in three ways that bear directly on the findings above. First, where prior emergency-call and crisis-text urgency classification work [8], [9], [10], [12] generally reports performance on a single corpus, this thesis's two-stage design surfaced a generalisation gap - and, more specifically, a modality-dependent reversal - that a single-corpus study structurally cannot detect. This is consistent with the out-of-event and out-of-type evaluation protocols used by Laurenti et al. [12] for crisis-tweet urgency detection, which found that performance also drops when a model must generalise to an unseen event, though that study did not separately evaluate acoustic evidence since it works with text-only social-media data.

Second, the multimodal design directly extends the speech-emotion-recognition literature's finding that acoustic cues carry information transcript content does not capture [4] into the emergency-call urgency setting specifically, where it had previously been underused. Third, the explicit treatment of Sinhala and Tamil context features separately, rather than relying on a general-purpose multilingual embedding, follows both the code-switching literature's finding that code-mixed text needs distinct modelling [3] and the practical limitation documented for Sinhala NLP tooling specifically - the absence of a mature Sinhala parser constraining what more sophisticated embedding techniques can currently achieve for this language [11]. No existing system identified in the literature review combines multimodal evidence, this specific multilingual code-mixed setting, and an independent authentic-data evaluation in the way this component does.

7.3 Research contribution

This thesis makes the following contributions to the research literature: (1) a systematic comparison of unimodal and multimodal (early- and late-fusion) urgency classifiers in a multilingual, code-mixed emergency-call setting, a combination not previously studied together; (2) an independent external validation against authentic DMC recordings, used directly to select the final fusion formulation rather than reported as a secondary check; (3) a quantitative domain-shift analysis, backed by a targeted augmentation experiment, identifying which feature families differ most between simulated and authentic data and showing that simple telephony-channel transformations do not by themselves explain that difference; and (4) a transparent, evidence-based account of exactly where multimodal fusion improves on text-only performance, and where it does not.

7.4 Product contribution

As a product component, this work contributes a working, reproducible urgency-classification service that can be integrated into the group's broader emergency-response system, together with a documented, version-controlled feature-extraction and classifier pipeline (Section 3.7) that the other components, and future development on this component, can build on. It also contributes a concrete, evidence-based recommendation for which classifier configuration to deploy - late fusion under the locked-ASR configuration - rather than the configuration that looked best on development data alone, which would have been the wrong choice for real calls.

7.5 Limitations and threats to validity

Both the simulated development corpus (267 calls) and, especially, the independent DMC validation set (59 calls) are modest in size for a three-class problem; DMC per-class recall estimates in particular carry substantial uncertainty given the small evaluation sample, and the 11-call fire/flood-scope-matched subset should be read as directional rather than precise. Because DMC results directly informed the choice of the final fusion formulation (Sections 3.2, 3.5), the generalisation estimate reported here reflects that external-validation role rather than an untouched, unbiased test set. The DMC set, while genuinely independent and authentic and spanning nine districts,

is drawn predominantly from flood-related operations; this thesis does not establish how well the component generalises to emergency types or regions materially different from that mix, and because the simulated training corpus was itself scoped to fire and flood scenarios, part of the DMC-Full versus scope-matched performance gap stems from that scope mismatch rather than from model or recognition failure.

Multilingual ASR quality, particularly for colloquial and code-mixed Sinhala/Tamil speech, remains an active source of error that feeds into every text/context feature, and confidence-weighting (Section 3.3) mitigates but does not eliminate that risk for the features it applies to. The domain-shift analysis identifies association, not full causal attribution; the augmentation experiment shows the specific noise/bandwidth transformations tested did not reduce the measured shift, but it does not isolate recording equipment, environment, or genuine behavioural differences from one another. Feature-selection decisions were made using the full development corpus prior to cross-validation (Section 3.5); that selection was targeted and evidence-driven for specific suspect features, plus block-level ablation for the acoustic family, rather than an exhaustive independent test of every individual feature. Formal user/stakeholder validation and full four-component integration testing, discussed in Sections 6.3 and 6.4, had not been completed at the time of writing, though informal feedback from the DMC Director General was positive and four-component integration testing is scheduled for the following week. No claim is made that this classifier offers a clinical or safety guarantee; it is reported strictly as a decision-support signal evaluated against the metrics presented.

7.6 Implications

The central implication of this thesis is methodological rather than purely technical: a classifier's apparent quality during development, measured however carefully, should not be trusted as an estimate of how it will behave on authentic data until that estimate has been checked against an independently sourced evaluation. For this component specifically, the implication is that deployment should proceed with the late-fusion, locked-ASR configuration rather than the early-fusion configuration that looked strongest during development, and that High-urgency recall - not raw accuracy - should be the metric decision-makers weight most heavily, given that under-flagging

a genuinely high-urgency call is plausibly a more costly error than over-flagging a lower-urgency one. More broadly, the finding that text-only features transfer at least as well as, and in places better than, the multimodal formulations implies that any future extension of this component should treat additional acoustic complexity as something to be justified by independent evidence, not assumed to help by default.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The value proposition of this component is to give emergency call-taking staff a consistent, evidence-based urgency signal that supplements - rather than replaces - their own judgement, particularly in the multilingual, code-mixed context of Sri Lankan emergency calls where transcript content alone is often an incomplete signal (Section 1.3). The primary target users are emergency-response call-taking and dispatch staff at agencies such as Sri Lanka's Disaster Management Centre, who must triage incoming reports under time pressure. Secondary users are the developers of the other three components in this group project, who consume this component's structured output as an input to the shared incident-fusion layer, and, more broadly, agency decision-makers responsible for allocating limited response resources.

8.2 Market and competitive position

As established in the literature and existing-systems review (Section 2.4), no identified existing system combines automatic urgency classification, multimodal (acoustic and textual) evidence, and support for Sri Lanka's specific multilingual, code-mixed setting. Most operational emergency-call handling in this context relies on human judgement, supported at most by call-routing and recording software rather than automatic classification. This component's competitive position is therefore not against other automated urgency classifiers in the Sri Lankan market, which do not appear to exist at the time of writing, but against the status quo of unassisted human triage; its relevant comparison point is whether it demonstrably improves on that status quo, which is precisely the question this thesis's independent DMC evaluation was designed to test.

8.3 Business or sustainability model

Given its origin in a research partnership with a government disaster-management agency, the most realistic path to sustained use for this component is institutional adoption by DMC or a similar public-sector emergency-response agency, rather than a direct-to-consumer or subscription commercial model. Sustainability under this model depends on continued maintenance - retraining or re-validating the pipeline as

more labelled authentic data becomes available, and keeping the ASR and diarization components current - which would most plausibly be supported through continued academic-institution or government partnership rather than commercial licensing revenue. A secondary, longer-term possibility, noted here as a direction rather than a committed plan, is adapting the component for other South Asian emergency-response contexts that share similar multilingual, code-mixed characteristics, where the underlying approach - rather than the specific Sinhala/Tamil feature lexicons - could transfer with re-labelling and re-validation.

8.4 Scalability intellectual property and compliance

Technically, the component is built on open-source tools with permissive licences (scikit-learn, librosa, pyannote.audio, and Whisper-family ASR models), so it carries no dependency on proprietary, closed-source components that would constrain future scaling or licensing decisions. Computationally, it is relatively inexpensive to scale: the classifiers themselves are Random Forests rather than large neural networks, and the main computational cost lies in automatic speech recognition and diarization, both of which are already required upstream regardless of this component's own processing. The frozen, version-controlled feature and model configuration (Section 3.7) is itself a scalability asset, since it makes retraining on a larger, future dataset a well-defined, reproducible operation rather than an ad hoc one.

Compliance considerations centre on the handling of real emergency call recordings and the personal, and sometimes distressing, content they may contain. This thesis's own data-handling practice, agreed with DMC and described in Section 3.6, would need to be formalised against Sri Lanka's applicable data-protection obligations before any operational deployment, and any future scaling to additional recording sources or agencies would need equivalent data-sharing and consent agreements rather than assuming the existing DMC arrangement extends automatically.

8.5 Contribution to the Sustainable Development Goals

This component's clearest alignment is with SDG 11 (Sustainable Cities and Communities), specifically target 11.5, which calls for significantly reducing the number of deaths and the economic losses caused by disasters, including a focus on

protecting the poor and people in vulnerable situations; faster, more consistent urgency triage of emergency calls is a direct, if modest, contribution toward that target. It also connects to SDG 9 (Industry, Innovation and Infrastructure), through the application of accessible, open-source machine-learning tooling to under-served multilingual NLP infrastructure for Sinhala and Tamil, and indirectly to SDG 3 (Good Health and Well-being), insofar as faster identification of high-urgency, life-threatening calls can shorten response times for medical and rescue assistance.

These contributions are stated as potential rather than demonstrated: this thesis evaluates the component's classification performance, not its real-world impact on response times or outcomes, and that impact would only be established through the operational deployment and stakeholder validation identified as future work in Chapter 9.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis set out to build and, critically, independently evaluate a multimodal urgency classifier for Sri Lankan multilingual emergency calls, treating that independent evaluation as central to the research rather than as a secondary confirmation. The central finding is that a substantial gap exists between simulated-corpus and authentic-recording performance, that this gap is not uniform across modalities, and that a domain-shift analysis - backed by a direct augmentation experiment showing that simple telephony-channel transformations do not by themselves explain the observed mismatch - provides quantitative, partial support for why. A late-fusion architecture, selected through this external validation rather than through development-set performance alone, produced substantially higher DMC High-urgency recall than the early-fusion formulation that had looked strongest during development, while text-only features remained a strong, and on some metrics superior, contributor even after fusion was improved.

The contribution of this work is not a claim of state-of-the-art or deployment-ready accuracy, but a methodologically transparent demonstration that simulated-corpus performance in this domain substantially overstates authentic-recording generalisation - and that this has to be measured and used to guide model and design decisions, rather than assumed.

9.2 Achievement of objectives

Revisiting the five objectives set out in Section 1.5: Objective 1 (designing an acoustic and textual feature set for the multilingual, code-mixed setting) was achieved, resulting in the 91-feature acoustic and 19-feature textual/conversational representations described in Chapter 3. Objective 2 (building and comparing unimodal and multimodal formulations on the simulated corpus) was achieved, with results reported in Section 6.1. Objective 3 (independently evaluating every formulation against authentic DMC recordings and using that evidence to select the final configuration) was achieved and is the central result of this thesis, reported in Sections 6.1 and 7.1 to 7.3. Objective 4 (quantifying domain shift and testing a simple explanation for it) was achieved through

the Cohen's d analysis and the audio-augmentation experiment reported in Section 6.1. Objective 5 (integrating the component's output with the other three components of the group project) was only partially achieved: the structured output format was designed for that purpose (Section 5.3), but end-to-end integration testing with the other components had not been completed at the time of writing, as reported honestly in Section 6.4.

9.3 Individual research and product contributions

The individual research contributions of this thesis are: a systematic comparison of unimodal and multimodal urgency classifiers in a multilingual, code-mixed emergency-call setting; an independent external validation against authentic DMC recordings used directly to select the final design; a quantitative domain-shift analysis identifying which feature families differ most between simulated and authentic data; and a transparent account of exactly where multimodal fusion helps and where it does not (Section 7.3). The individual product contributions are a working, reproducible urgency-classification pipeline with a version-controlled, documented configuration, and a concrete, evidence-based recommendation - late fusion under the locked-ASR configuration - for which design to carry forward into the group's integrated system, rather than the design that looked best on development data alone (Section 7.4).

9.4 Future directions

Several directions follow directly from the limitations identified in Section 7.5 and the gaps reported honestly in Chapter 6. In the near term: completing end-to-end integration testing with the AI agent, SMS intelligence, and multi-source incident-fusion components (Section 5.3, Section 6.4); conducting formal user or stakeholder validation with DMC call-taking staff, since the component's output has not yet been shown to a call-taker in a live or simulated dispatch setting (Section 6.3); and benchmarking the end-to-end pipeline's latency and throughput on production-representative hardware ahead of any deployment (Section 5.5).

In the medium term: growing the independent DMC corpus beyond 59 calls to tighten the statistical uncertainty on per-class estimates; investigating directly why early fusion fails to transfer to authentic recordings, to test the co-occurrence-pattern

explanation proposed but not isolated in Section 7.1; and working with DMC toward a more formally documented urgency taxonomy, to reduce the ambiguity in the Medium class identified both in the inter-annotator agreement result (Section 3.2) and in the modelling results (Section 6.1). Given this thesis's finding that text-only features are already highly competitive with, and in places superior to, the multimodal formulations, future acoustic feature work should be justified by independent DMC evidence before being adopted, rather than assumed to help by default. Finally, formally defining a cost matrix for the three urgency classes - rather than relying on the operational but informal judgement that under-flagging a High-urgency call is more costly than over-flagging a lower-urgency one (Section 1.7) - would allow future versions of this component to be tuned and evaluated against an explicit decision-theoretic objective rather than macro-F1 and recall alone.

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GLOSSARY

No additional glossary terms beyond those defined at first use in the text and listed in the List of Abbreviations.

APPENDICES

Appendix A Research instruments or extended results

Reproduction commands

The scripts below regenerate the results reported in this appendix and in Chapter 6, run from the project root. Each is deterministic given the frozen feature tables and fixed random seeds described in Section 3.7.

- python scripts/pipeline/02_extract_features.py (acoustic and textual feature extraction)
- python scripts/pipeline/03_train.py (locked classifier training on the simulated corpus)
- python scripts/analysis/06_final_candidate_and_tuning/results_table_with_baselines_and_ci.py (Figure 6.1, Figure 6.2)
- python scripts/analysis/04_domain_shift_and_scope/asr_source_comparison_late_fusion.py (Table A.1, Figure 6.3)
- python scripts/analysis/04_domain_shift_and_scope/domain_shift_diagnostic.py (Figure 6.4, Table A.3)
- python scripts/analysis/02_feature_ablation/acoustic_group_ablation.py (Table A.2)
- python scripts/analysis/01_error_diagnosis/dmc_error_analysis.py and medium_error_analysis.py (Section 6.1 error analysis)
- python scripts/analysis/06_final_candidate_and_tuning/verify_extraction_path.py (feature-extraction determinism check, Section 5.4).

This appendix provides the full confusion matrix underlying the summary discussed in Section 6.1 and Section 7.1, for the locked late-fusion model's predictions on the independent DMC Full evaluation set (59 calls), averaged over 5 reproducibility seeds (1, 7, 42, 99, 123) as reproduced from scripts/analysis/04_domain_shift_and_scope/asr_source_comparison_late_fusion.py. Rows are the true (DMC-assigned) label; columns are the model's predicted label; cell values are the mean number of calls across the 5 seeded re-fits, so they are not necessarily integers.

True \ Predicted	High	Low	Medium
High (n=23)	10.6	5.0	7.4
Low (n=22)	5.4	9.0	7.6
Medium (n=14)	3.4	4.4	6.2

Table A.1: Confusion matrix, locked late-fusion model, DMC Full (59 calls).

Precision, recall, and F1 per class, computed from this matrix: High 0.546 / 0.461 / 0.500; Medium 0.292 / 0.443 / 0.352; Low 0.489 / 0.409 / 0.446. The macro-F1 reported in Section 6.1 (0.428) is the mean of the five seeds' own per-seed macro-F1

values rather than a recomputation from this mean matrix, so it differs very slightly from macro-averaging the three F1 values above (0.433); both are reported here for transparency. The acoustic feature-subset ablation (Table A.2) and the extended domain-shift ranking (Table A.3) referenced in Section 3.5 and Section 6.1 are reported in full below. Per-seed simulated-corpus cross-validation outputs for every tested configuration, beyond the ablation summarised here, are retained in the project repository and are available on request, subject to the data-handling terms described in Section 3.6.

Table A.2 reports the block-level acoustic feature-subset ablation referenced in Section 3.5 and Section 4.3: each acoustic sub-group (MFCCs, pitch/prosody, energy, spectral) evaluated alone under the same stratified five-fold cross-validation as the main results, against the full 91-feature acoustic set. MFCCs alone (80 of the 91 acoustic features) reach macro-F1 0.611, close to the full acoustic set's 0.603, indicating that MFCCs carry most of the acoustic family's predictive value; pitch/prosody, energy, and spectral features individually reach only 0.542, 0.527, and 0.478 respectively, but their combination with MFCCs into the full 91-feature set still gives the best High-urgency recall (78.5%) of any acoustic sub-combination, which is why no sub-group was dropped from the production feature set despite MFCCs alone being nearly as strong on macro-F1.

Acoustic sub-group	Features	Macro-F1	High recall	Medium recall	Low recall
MFCCs only	80	0.611	76.9%	45.1%	63.6%
Pitch / prosody only	5	0.542	63.1%	43.9%	60.0%
Energy only	5	0.527	64.6%	36.6%	60.0%
Spectral only	1	0.478	57.7%	40.2%	49.1%
Full acoustic set (used)	91	0.603	78.5%	45.1%	60.0%

Table A.2: Acoustic feature-subset ablation, simulated corpus, 5-fold CV (stratified, 5-seed mean).

Table A.3 extends Figure 6.4 beyond its top 15 features, listing ranks 16-30 by |Cohen's d| between the simulated and DMC feature distributions. As in Figure 6.4, the extended

ranking remains almost entirely acoustic (MFCC coefficients and pitch), reinforcing that the measured domain shift is concentrated in the acoustic feature family rather than spread evenly across both modalities.

Rank	Feature	Family	Cohen's d
16	mfcc_7_std	acoustic	1.024
17	mfcc_39_std	acoustic	1.008
18	mfcc_12_mean	acoustic	0.978
19	mfcc_40_mean	acoustic	0.966
20	pitch_mean	acoustic	-0.956
21	mfcc_14_std	acoustic	0.942
22	mfcc_24_std	acoustic	-0.924
23	mfcc_29_std	acoustic	-0.871
24	mfcc_11_std	acoustic	0.865
25	mfcc_23_std	acoustic	-0.854
26	mfcc_26_mean	acoustic	-0.829
27	mfcc_39_mean	acoustic	0.824
28	mfcc_11_mean	acoustic	0.794
29	mfcc_6_mean	acoustic	-0.788
30	mfcc_30_std	acoustic	-0.788

Table A.3: Domain-shift ranking by |Cohen's d|, ranks 16-30 (continued from Figure 6.4).

Table A.4 reports the feature-selection ablation referenced in Section 3.5 and Section 4.3: successive development-corpus configurations, from acoustic-only through the addition and removal of specific text/context features, ending in the frozen 111-feature candidate used for every result reported in this thesis. Removing the per-language immediacy-count feature (D_context to D_no_immediacy) raised Medium recall from 43.9% to 46.3% with no change in High or further Low recall; adding the fire/smoke keyword-match indicator (D_no_immediacy to D_candidate, the frozen configuration) raised macro-F1 further and High recall from 75.4% to 79.2%, but did not itself change Medium recall further - the Medium-recall improvement is attributable specifically to immediacy removal, not to the fire/smoke indicator.

Configuration	Features	Macro-F1	High recall	Medium recall	Low recall
A_acoustic	91	0.592	76.2%	43.9%	60.0%
B_old_keyword	96	0.608	75.4%	43.9%	65.5%
C_base	96	0.593	76.2%	43.9%	60.0%
D_context	112	0.604	74.6%	43.9%	65.5%
E_context_gate	112	0.605	73.9%	45.1%	65.5%
D_no_immediacy	110	0.609	75.4%	46.3%	63.6%
D_candidate (frozen, used throughout)	111	0.621	79.2%	46.3%	63.6%

Table A.4: Feature-selection ablation, simulated development corpus (5-fold CV, pre-cross-validation development comparison, Section 3.5).

Worked example: one DMC call through the full pipeline

To make the abstract late-fusion architecture described in Section 4.2 and Section 4.4 concrete, this section traces one authentic DMC call end to end, using the locked late-fusion model (own fine-tuned Sinhala ASR, seed 42). The call (true label: High) was correctly classified, and is chosen specifically because its two branches disagree before fusion, illustrating why combining them is more than a formality.

Transcript (Sinhala ASR output, own fine-tuned model): "අනේ කුරුණෑයගෙලින් කටා කරන්නේ නුවර ඉන්වදුව කැම්පස්සෙකේ කර මැදබෝවලන් උසාවිය බිටි පස්ස්ක ගෙදරකින්නේ ජෙදර දැන් භාගයක් යටවෙලා ඒහරියට මොනුවායි හටයි එකයි දේශ්‍ය විසිසායි විසිහය යනුහට" (recording_id: "Call recording _251128_080730"). The feature extractor recorded keyword_count = 2 (the transcript matches terms in the flood-keyword lexicon, including යටවෙලා, "submerged/flooded under") and asr_quality = 1.0 (the transcript was not flagged as low-confidence).

Each branch was scored independently by its own frozen Random Forest, trained only on the simulated corpus: the text/context branch, seeing the flood-keyword match and context score, assigned P(High) = 0.783, P(Medium) = 0.214, P(Low) = 0.003 - a confident High prediction. The acoustic branch, seeing only the prosodic and spectral features extracted from the caller's voice, disagreed: P(High) = 0.275, P(Medium) =

0.513, P(Low) = 0.212 - its own highest-probability class was Medium, not High. Taken alone, the acoustic-only model would have misclassified this call.

Late fusion combines the two branches by unweighted averaging: P(High) = 0.5(0.783) + 0.5(0.275) = 0.529, P(Medium) = 0.5(0.214) + 0.5(0.513) = 0.363, P(Low) = 0.5(0.003) + 0.5(0.212) = 0.107. High remains the highest-probability class after fusion, so the final prediction is High - correctly matching the true label. This is precisely the operational case the architecture is designed for: a genuine disagreement between modalities, resolved correctly because the text branch's confident, keyword-grounded signal was strong enough to outweigh the acoustic branch's comparatively weaker and incorrect lean toward Medium, without either branch being allowed to dominate unconditionally. Section 6.1's error analysis shows this does not always happen in the model's favour - the 10 MODEL_OVERRIDE errors reported there are cases where the correct branch's signal was instead outweighed by the incorrect one - but this example demonstrates the mechanism working as intended on a real, authentic recording.

Appendix B Academic integrity and AI use disclosure

Tools used. Claude (Anthropic), a generative AI assistant, via the Claude desktop application.

Purpose and sections affected. Drafting support for the prose of this thesis and the related ICAC 2026 conference paper, from the author’s own codebase and experiment outputs; code review and tracing of numerical claims back to their source scripts; verification of external citations against the original papers before use; and assistance reconstructing draft log-book entries from the project’s file and commit history for the author to review. Generative AI assistance contributed to drafting across Chapters 1 through 9 of this thesis.

Verification. Every numerical claim, methodological description, and citation was checked by the author against the underlying code, experiment outputs, or original source paper; two factual corrections were identified and made during this process. The author reviewed and remains fully responsible for the accuracy, originality, and correctness of all content, code, data, analyses, and conclusions in this thesis.